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United States Patent	12405266
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Wachernig; Hanno

Method for determining the number of antibodies on exosomes using quality control according to a first method of design recognition and according to a second method of electrophoresis

Abstract

The invention relates to a method for determining the florescence of antibodies on exosomes using quality control of the measurement using two different control methods.

Inventors:	Wachernig; Hanno (Diessen a. A., DE)
Applicant:	PARTICLE METRIX GMBH (Inning am Ammersee, DE)
Family ID:	70802553
Assignee:	PARTICLE METRIX GMBH (Inning am Ammersee, DE)
Appl. No.:	17/607988
Filed (or PCT Filed):	April 27, 2020
PCT No.:	PCT/DE2020/000083
PCT Pub. No.:	WO2020/221382
PCT Pub. Date:	November 05, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20220341917 A1	Oct. 27, 2022

Foreign Application Priority Data

DE	10 2019 003 132.6	May. 02, 2019
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Publication Classification

Int. Cl.: G01N33/50 (20060101); G01N21/64 (20060101); G01N33/53 (20060101)

U.S. Cl.:

CPC G01N33/5076 (20130101); G01N21/6456 (20130101); G01N33/5308 (20130101);

Field of Classification Search

CPC: G01N (15/01); G01N (15/06); G01N (15/075); G01N (2015/0038); G01N (21/6456); G01N (33/5076); G01N (33/5308)

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Primary Examiner: Kaur; Gurpreet

Attorney, Agent or Firm: Duane Morris LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a § 371 national stage entry of International Application No. PCT/DE2020/000083, filed on Apr. 27, 2020, which claims priority to German Patent Application No. 10 2019 003 132.6, filed on May 2, 2019, the entire contents of which are incorporated herein by reference.

(2) The application relates to a method for determining the fluorescence and the number of antibodies on exosomes using a quality control by a first method of pattern recognition and by a second method of electrophoresis.

(3) In biology, vesicles are very small, round to oval, intracellular bubbles which are surrounded by a monolayer or bilayer membrane or a reticulated shell of proteins. Vesicles form their own cell compartment, in which different cellular processes take place. Their size is about one micrometer. Vesicles are responsible for the transport of many substances in the cell.

(4) The mechanisms which lead to the formation of extracellular vesicles have not yet been fully clarified.

(5) Here, exosomes are small vesicles having a size of approx. 50 to 150 nm.

(6) Extracellular vesicles play a major role in intercellular communication in healthy tissue, but also in various diseases. They are of interest to various therapeutic methods.

(7) In short: exosomes are extracellular microvesicles which are released by living cells by means of exocytosis, which contain a multitude of proteins and RNA and which exercise biological functions.

(8) The sole, comprehensive analysis method which leaves exosomes in their natural environment and is capable of evaluating particles with dimensions as small as 20 nm for all possible measurement parameters is NTA or nanoparticle tracking analysis. Said analysis method therefore also has the potential of being able to diagnose the different proteins situated on the surface of the membrane of the exosomes. The detection of different proteins on the surface of the exosomes is achieved using specific molecules known from the immune system. These are antibodies.

(9) Antibodies are highly specific and recognize and bind exclusively to a certain protein.

(10) Thus, if what is wanted is to know whether a certain protein X is present on the surface of an exosome, use is made of an antibody known to specifically recognize the protein X and dock therewith. If the exosome has many of such proteins X, all of said proteins can, in the best case, be recognized and bound by the specific antibodies. If said antibodies have been artificially provided with a fluorescent dye, the entire complex consisting of exosome/antibody/fluorescent dye can be detected by the Zetaview instrument.

(11) If another protein Y is of interest, use is accordingly made of an antibody Y which exclusively recognizes and binds protein Y in a specific manner. If said antibody Y has also been labeled with a

fluorescent dye, this exosome/antibody Y/fluorescent dye complex can likewise be visualized in a fluorescence channel. Now, if the Zetaview recognizes a signal in a fluorescence channel, it can be assumed that the relevant exosome bears the protein of interest on its surface. If no fluorescence signal is visible, the relevant protein does not exist on the surface of the exosome.

(12) Using this apparatus, it is possible, then, to carry out so-called colocalization experiments.

(13) Colocalization means that two or more different proteins are represented side by side on the surface of the exosomes. In our example, these would be protein X and protein Y. What we would thus like to know is whether these two proteins exist side by side on the exosome surface. To this end, it is possible, for example, to label antibody X with a red fluorescent dye and antibody Y with a blue fluorescent dye.

(14) If both labeled antibodies recognize and bind the relevant proteins on the exosome surface, or are represented (colocalized), signals are captured in both fluorescence channels of the ZetaView instrument. If only protein X is represented, a signal is detected only in the red fluorescence channel (because only antibody X can bind), but not in the blue fluorescence channel (and vice versa).

(15) It is thus possible using just a single measurement and a single sample to find out whether different proteins of interest are simultaneously represented (colocalized) on the surface of the exosomes.

(16) In relation to the prior art, what follows is a description of the colocalization analysis of two antibodies on the same exosome by means of an NTA.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a particular NTA nanoparticle analysis.

(2) FIG. 2 depicts a change wheel for a color filter 17.

(3) FIG. 3 shows the course of an analysis, multiple lasers and multiple color filters.

(4) FIG. 4a shows a Brownian motions of particles.

(5) FIG. 4b shows size distribution of the two groups of particles in a bar chart.

(6) FIG. 4c shows a scatter plot from the pattern analysis, depicting the calculated particle size distributions of the two patterns.

(7) FIG. 4d shows the temporal course of a video recording of a particle of group G1.

(8) FIG. 4e shows the temporal course of a video recording of a particle of group G2.

(9) FIG. 5 depicts an electrophoresis and electroosmosis measurement in the measurement cell for comparison of the measurement from the prior art.

(10) FIG. 6 depicts the cumulative electrophoretic mobility distribution curves of two samples.

DETAILED DESCRIPTION OF THE INVENTION

(11) FIG. 1 depicts an NTA nanoparticle-tracking measurement arrangement from Particle Metrix that allows the detection of multiple antibodies on the same exosome. For this purpose, it is necessary for multiple combinations of lasers and filters to follow one another in a rapid manner. Since the nanoparticles carry out Brownian motion, they can disappear from the video image after a few seconds, meaning that there are only a few seconds for simultaneous measurement of multiple fluorescences on the very same particle, though in vitro. This means that the laser/filter combination has to change within fractions of a second. Since it is nowadays already possible to guide multiple lasers onto a fiber and to change rapidly from one wavelength to the next by electronic switching, this is no longer a problem. What only remains is to accomplish rapid filter change and to carry out analysis measurement rapidly (within 1 to 10 seconds).

(12) In this form of NTA, the optical system to be imaged consists of an objective having an optionally additional liquid lens 6, which objective is directed to the particulate sample 9 in the

measurement cell **10**, and a video camera **15** or detector which records the movement of the particles.

(13) The lasers **1** to **4** shown here, as a light source which is intermittently switched via the collection prisms **14** and the beam path **21**, serve as an excitatory light source for scattered light and fluorescence light and radiate into the laser pass-through window **22** of the measurement cell **10** at an angle of 90 degrees.

(14) At the camera **15**, what is simultaneously registered by means of the control system of the optical system **18** is light from the fluorescence plane **5** and the scattered light plane **8** via the convergent beam path through a combination of microscope objective (not indicated) and liquid lens **6** with adjustable focus. The touchscreen display **19** and the control system containing the particle tracking program **20** are connected to the video camera **15**.

(15) In the convergent beam path, in the transverse direction, a change wheel **17** for color filters is arranged on a filter wheel axle **11** and a counterbearing **7** over a change coupling **12** and a filter drive **13**, the change wheel being fitted with color filters **16** and the color filter wheel being controlled by means of a device **26**. The light which passes through the optical-system pass-through window **23** of the measurement cell **10** and the color filter **16** is guided, via the combination of microscope objective and liquid lens **6**, into the video camera **15**, where it is registered.

(16) FIG. **2** depicts the change wheel for the color filter **17**.

(17) In the center, what can be seen is the filter wheel axle **11** surrounded by the filters **25**, **26**, **27**, **28** and **29**. Here, the filter **25** in the form of a quartz glass insert can be considered neutral, whereas the filter numbered **26** opens at a wavelength above 420 nm. The filter numbered **27** opens at a wavelength above 500 nm, the filter **28** opens at a wavelength above 540 nm and the filter **29** opens at 660 nm.

(18) FIG. **3** shows the position of the particular color filter for fluorescence emission measurement and of the particular laser in the course of an analysis.

(19) On the left-hand side of FIG. **3**, six positions of the color filter **16** are shown by way of example, and at the uppermost position the respectively engaged color filter is indicated by its number and the respectively engaged laser is indicated.

(20) The method of the technique from the prior art shown in FIGS. **1** to **3** can be compared with **2** further methods in order to thus obtain a statement about the quality of the measurements.

(21) It is an object of the present invention to obtain an easy-to-obtain quality control for the described prior art. This is achieved by the method according to claim **1**:

(22) a method for determining the fluorescence and the number of antibodies on exosomes using a measurement quality control, comprising the following features: a) for preparation for a measurement, a measurement cell (**10**) is filled with a solution composed of fluorescent antibodies and exosomes as sample (**9**), b) a combination of lasers and filters is used for irradiation of the measurement cell (**10**) in vitro with rapid changeover and an optical device (**15**) is used for imaging thereof, c) the overall control system for the measurement arrangement that is used is a Particle Metrix measurement arrangement, d) a control measurement establishes pattern differences between exosomes without antibody conjugation and those with antibody conjugation, e) a further control measurement establishes differences in relation to the prior art by means of an electrophoretic analysis.

(23) Using one instrument and the same measurement cell, the samples from the technique from the prior art are analyzed and compared in two different methods, by the method of pattern recognition and by the method of electrophoresis.

(24) In what follows, the invention will be described in more detail.

(25) In the figures:

(26) FIG. **4**: shows graphs in the monitor **19** due to camera recording and pixel analysis from the pattern recognition for comparison of the measurement from FIG. **1**

(27) FIG. 5: depicts an electrophoresis and electroosmosis measurement in the measurement cell for comparison of the measurement from the prior art

(28) FIG. 6: depicts the cumulative electrophoretic mobility distribution curves of two samples

(29) FIG. 4

(30) shows that subgroups of particles (exosomes) can be depicted differently on video. What are meant are groups of exosomes with docked antibodies (proteins) or exosomes without antibodies. In the case of scattered light measurement and of fluorescence light measurement, the subgroups can be calculated by video pattern recognition and presented in graphs. The pattern recognition analysis can be carried out in the scattered light measurement and the fluorescence light measurement and is used for quality control of the technique from the prior art.

(31) FIG. 4a shows a video of the Brownian motion of the particles, from which the size follows, the mean square displacement ($\Delta x \cdot \Delta y$) of an individual particle **44** the starting position (**1a**) and the end position (**2a**)—measured in one second and inserted into Stokes' Einstein formula—yields the particle size. The video recording contains additionally further information that is analyzed and evaluated by the pattern recognition program.

(32) These include the brightness, shape, temporal behavior and trajectory lengths of a particle.

(33) FIG. 4c shows a scatter plot from the pattern analysis, depicting the calculated particle size distributions of the two patterns.

(34) The example shows the size distribution of exosomes of a HEK293 cell culture, with simultaneous plotting of the temporal behavior of the intensity of each particle (mean intensity) against the size. In the scatter plot, two groups of particles can be seen. Group G1 corresponds to the particle features as depicted in FIG. 4d. Group G2 corresponds to the particle features as depicted in FIG. 4e (non-exosomes).

(35) From the comparison of group G1 of exosomes measured before the conjugation with antibodies and afterwards, it is possible to deduce pattern differences. As a result, it is possible by means of video pattern recognition to quantify the antibodies on an exosome, i.e., the number of antibodies which have docked with an exosome.

(36) FIG. 4d shows the temporal course of a video recording of a particle of group G1.

(37) The particle is small, less bright and its intensity looks uniform over the course of time.

(38) Exosomes are concerned here, since they have a membrane shell, have approximately spherical symmetry and do not flicker.

(39) FIG. 4e shows the temporal course of a video recording of a particle of group G2.

(40) The particle is larger and its light flickers between very bright and very dark.

(41) In the example, the “flickering” particles can be excluded as non-exosomes. FIG. 4b shows the size distribution of the two groups of particles in a bar chart. What can be seen here are group G1 as exosomes (115 nm) and group G2 as non-exosomes (247 nm).

(42) The pattern recognition distinguishes the exosomes from other particles and calculates their number in the sample, as shown in FIG. 4c and FIG. 4d. Following a measurement before and after conjugation, the exosomes can be calculated in terms of their size, as shown in FIG. 4a. The difference between the two measurements provides information as to how many antibodies have docked with an exosome.

(43) The result is used for checking of the measurement from technique (prior art) shown in FIG. 1 to FIG. 3.

(44) FIG. 5 shows the schematic depiction of an electrophoretic analysis for particles as a second comparative measurement for the technique from the prior art.

(45) From what is depicted by FIG. 5, it is clear that electroosmosis is triggered by an applied field **41**, generated by the plus electrode **30** and minus electrode **31**. Its course of movement is depicted by the curve **36**.

(46) The electrodes **30**, **31** consist of highly conductive graphene in order to produce optimal contact with the suspension in the measurement cell. This avoids interfering bubble formation and

the movements of the particles are more homogeneous.

(47) If charged particles are present in the liquid, they move at a velocity v in the electric field E **41**. This effect is even across the entire cross section of the cell and is called electrophoresis. What is measured is the electrophoretic mobility $\mu_{sub.e} = v/E$. Electrophoresis and electroosmosis are independent of one another and add up. The magnitude of the addition is depicted by three arrows **40**. The curve **37** shows the course of movement of electroosmosis and electrophoresis $\mu_{sub.e}$ together in the field **41**. The dimension of mobility is reported in $\mu\text{m/s/V/cm}$. In the case of very high loads, the value displayed by the mobility can be up to $5 \mu\text{m/s/V/cm}$.

(48) If what is wanted is to measure the electrophoresis $\mu_{sub.e}$ independently of the electroosmosis, this must be done in the so-called stationary layers **34** and **35** (SL) of the channel of the measurement cell. At the layers **34**, **35**, the electroosmosis reverses and is therefore zero. Therefore, at these two layers, it is the electrophoresis alone that is readable. From the geometry of the measurement cell, it is possible to deduce the stationary layers of SL1 **34** and SL2 **35**. The laser radiates into the measurement cell at an angle of 90 degrees to the microscope plane and perpendicularly through the window **22**. For measurement, the laser focus **21** and the microscope focus **38**, **39** are synchronously alternately directed to SL1, **34** and SL2, **35**.

(49) If only the comparisons of the electrophoretic properties, i.e., differences in the ionic loading of the particle surfaces, need to be examined, they can be measured at any desired layer of the measurement cell **10** (along the cell walls). This means that comparisons of the electrophoretic mobility $\mu_{sub.e}$ can be made at any desired point of the measurement cell, provided that the distance from the cell walls does not change during the comparative measurement. In order to compare pure exosomes and antibody-conjugated exosomes with one another, the ionic conditions of the surrounding liquid (salt type and conductivity) must be identical.

(50) The procedure for a comparative measurement is implemented in 7 steps.

(51) A sample **1+** (without antibody) is compared with a sample **2** (with antibody).

(52) Steps 1-3 is the calibration (sample **1**) of the starting point of the measurement.

(53) 1st Step:

(54) In a vessel, a certain amount of a suitable liquid is mixed with a certain amount of neutral exosomes.

(55) 2nd Step:

(56) In the measurement cell **10**, the electrophoretic mobility of sample **1** is measured and a mean value, the EM **50**, is calculated. The EM **50(1)** is described in more detail in FIG. **6**.

(57) 3rd Step:

(58) An exact amount of a cationic polymer (reagent) is added to sample **1** and the EM **50** is ascertained as in step 2.

(59) The EM **50(1)** is our constant value depicted as curve **37** in FIG. **5** and as curve **1** in FIG. **6**. Said value is saved for all further experiments with the suspension of sample **1+**. The calibration is completed.

(60) 4th Step:

(61) A certain amount containing different fluorescent antibodies is added to a certain amount of the calibrated exosome suspension (sample **1+**) and conjugation is measured in the measurement cell **10** by technique **1** (prior art).

(62) 5th Step:

(63) At the same time, the EM **50(2)** can be ascertained in the measurement cell using the sample from step 4 (exosomes+antibody). This is depicted by curve **42** in FIG. **5** and by curve **2** in FIG. **6**.

(64) 6th Step:

(65) The value from step 3 (EM **50(1)** calibration) is compared with the value from step 5 (EM **50(2)** exosomes+antibody) and the number of antibodies docked on an exosome is calculated.

(66) 7th Step:

(67) Comparison of the Measurement Values

(68) By comparing the value from the technique according to the prior art (step 5) and the value from step 6, it is possible to check the quality of the measurement from the prior art.

(69) FIG. 6 shows a graph for the cumulative electrophoretic mobility distribution in % for easier readability of differences in the charge states described by the EM 50(1) of sample 1+ and the EM 50(2) of sample 2 (exosomes+antibody). What is shown is the change in specific surface charge Q of an exosome upon conjugation with an antibody per surface area F [$C/nm.sup.2$]. More clearly, how many elementary charges x per surface element [$\#e/nm.sup.2$] are lost from the exosomes by binding per μg of antibodies.

(70) The analysis requires complex control of the described method sequences and movement sequences by a specific control and analysis program.

LIST OF REFERENCE SIGNS

(71) **1** Laser (violet=405 nm) **2** Laser (blue=488 nm) **3** Laser (green=520 nm) **4** Laser (red=640 nm) **5** Fluorescence plane **6** Liquid lens with adjustable focus **7** Counterbearing of the filter wheel axle **8** Scattered light plane **9** Sample **10** Measurement cell **11** Filter wheel axle **12** Change coupling **13** Filter wheel drive (stepper motor) **14** Collection prisms (all guide lasers into a beam path) **15** Detector or video camera (CCD, CMOS) **16** Color filter (edge filter or bandpass filter) **17** Change wheel for color filters **18** Control system for optical system **19** Touchscreen display **20** Overall control system containing particle tracking program **21** Beam path for laser **22** Laser pass-through window of the measurement cell **10** **23** Optical-system pass-through window of the measurement cell **10** **24** Control system for color filter wheel **25** Filter, neutral (quartz glass) **26** Filter, opens above 420 nm **27** Filter, opens above 500 nm **28** Filter, opens above 540 nm **29** Filter, opens above 660 nm **30** Field electrode + **31** Field electrode – **32** Counterion layer **33** Inner glass wall of measurement cell (– negatively charged) **34** Stationary layer 1 (SL 1) **35** Stationary layer 2 (SL 2) **36** Curve of purely electroosmotic movement (particles are not charged) **37** Curve of electroosmotic and electrophoretic movement (particles are positively charged), exosomes without antibody (quicker) **38** Focal plane of layer 1 (**34**) **39** Focal plane of layer 2 (**35**) **40** Magnitude of electrophoresis v/E **41** Electric field (direction of movement) **42** Curve of electroosmotic and electrophoretic movement, exosomes with antibody (slower) **43** Magnitude between exosomes with and without antibody **44** Particle

Claims

1. A method for determining the fluorescence and the number of antibodies on exosomes using a measurement quality control, comprising the following features: a) for preparation for a measurement, filling a measurement cell (**10**) with a solution composed of fluorescent antibodies and exosomes as sample (**9**), b) performing a first measurement with a nanoparticle tracking measurement arrangement including a combination of lasers and filters for irradiation of the measurement cell (**10**) in vitro with rapid changeover and an optical device (**15**) for imaging thereof, c) analyzing images detected by the optical device (**15**) in the first measurement for detecting the fluorescence and a number of antibodies by nanoparticle tracking analysis, d) performing a first comparative measurement by video pattern recognition using the sample, the measurement cell, and the nanoparticle tracking measurement arrangement as used in the first measurement of step b), wherein pattern differences between exosomes without antibody conjugation and those with antibody conjugation are detected, and e) performing a video pattern recognition analysis for said first comparative measurement for detecting the number of antibodies, and comparing the number of antibodies detected in the first measurement of step b) with the number of antibodies detected in the first comparative measurement for quality control.

2. The method according to claim 1, further comprising: performing a further comparative measurement by means of liquid electrophoresis using the sample, the measurement cell and the nanoparticle tracking measurement arrangement used in the first measurement of step b);

- performing an electrophoretic analysis for the further comparative measurement for detecting the number of antibodies, and comparing the number of antibodies detected in the first measurement of step b) and the number of antibodies of the further comparative measurement for quality control.
3. The method as claimed in claim 2, characterized that in the liquid electrophoresis, graphene which is highly conductive for the electrodes is used when choosing the electrodes.
 4. The method as claimed in claim 2, wherein, for the liquid electrophoresis: a calibration for detecting a first mean value (EM 50(1)) for an exosome suspension without antibodies is performed, antibodies are added to the calibrated exosome suspension and a second mean value (EM 50(2)) of the resulting exosome suspension is detected, and a number of antibodies is calculated from the first mean value (EM 50(1)) and the second mean value (EM 50(2)).
 5. A computer program containing non-transitory program code for carrying out the method steps as claimed in claim 1 when the program is executed in a computer.
 6. A computer program containing non-transitory program code for carrying out the method steps as claimed in claim 2 when the program is executed in a computer.
 7. A computer program containing non-transitory program code for carrying out the method steps as claimed in claim 3 when the program is executed in a computer.
 8. A computer program containing non-transitory program code for carrying out the method steps as claimed in claim 4 when the program is executed in a computer.
 9. A machine-readable medium containing the non-transitory program code of a computer program for carrying out the method as claimed in claim 1 when the method is executed in a computer.
 10. A machine-readable medium containing the non-transitory program code of a computer program for carrying out the method as claimed in claim 2 when the method is executed in a computer.
 11. A machine-readable medium containing the non-transitory program code of a computer program for carrying out the method as claimed in claim 3 when the method is executed in a computer.
 12. A machine-readable medium containing the non-transitory program code of a computer program for carrying out the method as claimed in claim 4 when the method is executed in a computer.
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